Integrating Factors

- Many first-order equations $M\,dx + N\,dy = 0$ are *not* exact as they stand ($M_y \neq N_x$).
- Often, however, multiplying the entire equation by a suitably chosen function $\mu(x, y)$ makes it exact. Such a function is called an **integrating factor**.
- A function $\mu(x, y) \neq 0$ is called an **integrating factor** for $M\,dx + N\,dy = 0$ if the equation $\mu M\,dx + \mu N\,dy = 0$ is exact.
- In this case,
$$\frac{\partial}{\partial y}(\mu M) = \frac{\partial}{\partial x}(\mu N).$$
- Expanding this condition gives the **integrating factor PDE**:
$$\mu_y M - \mu_x N = \mu\,(N_x - M_y)\,. \tag{1}$$



IF - Computation

- In general the partial differential equation (1) for μ, harder to solve than the original ODE.
- However, in many practical cases μ can be assumed to depend on only one variable (x alone, y alone, or a simple combination such as xy or $x+y$), which reduces the PDE to an ordinary differential equation for μ.
- Case 1: μ is a function of x alone:
- In this case $\mu_y = 0$.
- Thus from (1), $\frac{1}{\mu}\frac{d\mu}{dx} = \frac{M_y - N_x}{N}$.
- Since the LHS is a function of x only, then we need the RHS $\frac{M_y - N_x}{N} := f(x)$ must be a function of x only.
- Integrating the above equation, we obtain the integrating factor as

$$\mu(x) = \exp\left(\int f(x)\,dx\right) = \exp\left(\int \frac{M_y - N_x}{N}\,dx\right).$$



IF - Computation

- Case 2: μ is a function of y alone
- In this case $\mu_x = 0$.
- Thus from (1), $\frac{1}{\mu}\frac{d\mu}{dy} = -\frac{M_y - N_x}{M}$.
- Since the LHS is a function of y only, then we need the RHS $-\frac{M_y - N_x}{M} := g(y)$ must be a function of y only.
- Integrating the above equation, we obtain the integrating factor as

$$\mu(y) = \exp\left(\int g(y)\, dy\right) = \exp\left(\int \frac{N_x - M_y}{M}\, dy\right).$$



IF - Computation

- Case 3: μ is a function of the product xy
- Since $\mu = \mu(z)$ where $z = xy$, using it in (1), we obtain

$$\mu_z xM - \mu_z yN = \mu(N_x - M_y) \implies \frac{1}{\mu}\frac{d\mu}{dz} = \frac{M_y - N_x}{yN - xM}.$$

- Since the LHS is a function of z only, then we need the RHS $\frac{M_y - N_x}{yN - xM} := h(z)$ must be a function of z only.
- Integrating the above equation, we obtain the integrating factor as

$$\mu(z) = \exp\left(\int h(z)\, dz\right) = \exp\left(\int \frac{M_y - N_x}{yN - xM}\, dz\right), \quad z = xy.$$



Example - Case 1

- Solve $(3xy + y^2)\,dx + (x^2 + xy)\,dy = 0$.
- Here $M = 3xy + y^2$, $N = x^2 + xy$ gives that $M_y = 3x + 2y, \qquad N_x = 2x + y \quad \Rightarrow \quad M_y \neq N_x$, so the equation is not exact.
- Observe that $\frac{M_y - N_x}{N} = \frac{(3x+2y)-(2x+y)}{x^2+xy} = \frac{x+y}{x(x+y)} = \frac{1}{x}$, which depends only on x.
- Hence $\mu(x) = \exp\left(\int \frac{1}{x}\,dx\right) = e^{\ln x} = x$.
- Multiplying through by $\mu = x$ in given equation, we obtain

$$(3x^2y + xy^2)\,dx + (x^3 + x^2y)\,dy = 0.$$

- Note that $\partial_y(3x^2y + xy^2) = 3x^2 + 2xy = \partial_x(x^3 + x^2y)$.
- Therefore, there exists a function $F(x, y)$ such that

$$F = \int (3x^2y + xy^2)\,dx = x^3y + \tfrac{1}{2}x^2y^2 + g(y),$$

$$F_y = x^3 + x^2y + g'(y) = x^3 + x^2y \quad \Rightarrow \quad g'(y) = 0.$$

- General solution: $F(x, y) \equiv x^3y + \frac{1}{2}x^2y^2 = C$.



Example - Reducible to Exact

- Solve $(xy-1)\,dx+(x^2-xy)\,dy=0$.
- Here $M=xy-1$, $N=x^2-xy$ and
  $M_y=x, \qquad N_x=2x-y \quad \Rightarrow \quad M_y\neq N_x.$
- Observe that $\frac{N_x-M_y}{M}=\frac{(2x-y)-x}{xy-1}=\frac{x-y}{xy-1}$ is is not a function of y alone.
- Again, we need to check $\frac{M_y-N_x}{N}=\frac{x-(2x-y)}{x^2-xy}=\frac{y-x}{x(x-y)}=-\frac{1}{x}$ as a function of x only.
- Thus, we get IF as $\mu(x)=\exp\left(\int -\frac{1}{x}\,dx\right)=e^{-\ln x}=\frac{1}{x}$.
- Multiplying given DE through by $\mu=1/x$, we obtian an exact DE $\left(y-\frac{1}{x}\right)dx+(x-y)\,dy=0$.
- By standard procedure,

$$F=\int\left(y-\frac{1}{x}\right)dx=xy-\ln|x|+g(y), \qquad F_y=x+g'(y)=x-y$$

- General solution: $F(x,y)\equiv xy-\ln|x|-\frac{y^2}{2}=C$.



Example - Reducible to Exact

- $y\,dx + (2x - ye^y)\,dy = 0.$
- Here $M = y, \qquad N = 2x - ye^y$ and
  $M_y = 1, \quad N_x = 2 \quad \Rightarrow \quad$ Not exact.
- Observe that $\frac{N_x - M_y}{M} = \frac{2-1}{y} = \frac{1}{y}$ (function of y only).
- Then, the IF is given by $\mu(y) = e^{\int \frac{1}{y}\,dy} = e^{\ln y} = y.$
- Multiply through by $\mu = y$, we have exact DE
  $y^2\,dx + (2xy - y^2e^y)\,dy = 0.$
- Since it is exact, there exists F such that
  $F(x, y) = \int y^2\,dx = xy^2 + g(y).$
- Again, $F_y = 2xy + g'(y) = 2xy - y^2e^y \;\Rightarrow\; g'(y) = -y^2e^y$
- Integrating by parts, we have
  $g(y) = -\int y^2e^y\,dy = -e^y(y^2 - 2y + 2).$
- The general solution is given by $\boxed{xy^2 - e^y(y^2 - 2y + 2) = C}$.



Example - Reducible to Exact

- Solve $x\,dy + y\,dx + 3x^3y^4\,dy = 0$.
- Rewriting it as $\underbrace{y}_{M}\,dx + \underbrace{(x + 3x^3y^4)}_{N}\,dy = 0$.
- Since $M_y = 1$ and $N_x = 1 + 9x^2y^4$ are not equal, the equation is not exact.
- Note that $N_x - M_y = (1 + 9x^2y^4) - 1 = 9x^2y^4$ does not help to use Case 1 and Case 2 for the computation of IF.
- But $\frac{N_x - M_y}{xM - yN} = \frac{9x^2y^4}{-3x^3y^5} = -\frac{3}{xy} = -\frac{3}{z}$, $z = xy$.
- For IF, $\frac{d\mu}{\mu} = -\frac{3}{z}\,dz \implies \ln\mu = -3\ln z \implies \mu = z^{-3}$.
- Multiply IF with the given equation, we have the exact DE $\frac{y}{x^3y^3}\,dx + \frac{x + 3x^3y^4}{x^3y^3}\,dy = 0$.
- Then, $F(x,y) = \int \frac{1}{x^3y^2}\,dx = -\frac{1}{2x^2y^2} + g(y)$ and further computations give $g(y) = \frac{3}{2}y^2$ (HOMEWORK).
- The general solution is $\boxed{3x^2y^4 - Cx^2y^2 = 1}$.



Results

- If

$$\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N - M}$$

is a function of $z = x + y$, then

$$e^{\int \frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N-M}\, dz}$$

is an IF of $M\,dx + N\,dy = 0$. **Homework**

- If the IF of $M\,dx + N\,dy = 0$ is a function of $z = x - y$, then compute the formula for IF. **Homework**
- If $M\,dx + N\,dy = 0$ is homogeneous and $Mx + Ny \neq 0$, then

$$\frac{1}{Mx + Ny}$$

is an IF.



Sketch of Proof in Homogeneous Case

$$M\,dx + N\,dy = \frac{1}{2}\left[(Mx+Ny)\left(\frac{dx}{x}+\frac{dy}{y}\right)+(Mx-Ny)\left(\frac{dx}{x}-\frac{dy}{y}\right)\right]$$

$$= \frac{1}{2}\left[(Mx+Ny)\,d(\ln xy)+(Mx-Ny)\,d\left(\ln\frac{x}{y}\right)\right] \tag{2}$$

$$\Rightarrow \quad \frac{M\,dx+N\,dy}{Mx+Ny} = \frac{1}{2}\left[d(\ln xy)+\frac{Mx-Ny}{Mx+Ny}\,d\left(\ln\frac{x}{y}\right)\right]$$

$$= \frac{1}{2}\,d(\ln xy)+\frac{1}{2}\,f\left(\frac{x}{y}\right)d\left(\ln\frac{x}{y}\right) \qquad \text{(say)}$$

$\left(\because M\,dx+N\,dy \text{ is homogeneous} \Rightarrow \frac{Mx-Ny}{Mx+Ny} \text{ is also homogeneous,}\right.$
and we can write it as $f\left(\frac{x}{y}\right)\Big)\ \left(\because \frac{x}{y} = e^{\ln\frac{x}{y}}\right)$

$= \frac{1}{2}\,d(\ln xy)+\frac{1}{2}\,F\left[\ln\left(\frac{x}{y}\right)\right]d\left(\ln\frac{x}{y}\right)$ which is an exact differential.



Example - Reducible to Exact

- Solve $(x^2y - 2xy^2)\,dx + (3x^2y - x^3)\,dy = 0$.
- Here $M = x^2y - 2xy^2, \quad N = 3x^2y - x^3$ and $M_y = x^2 - 4xy \neq N_x = 6xy - 3x^2$.
- The given DE is **not exact**. But it is a homogeneous function of degree 3.
- $\therefore \quad \frac{1}{Mx+Ny} = \frac{1}{(x^2y-2xy^2)x+(3x^2y-x^3)y} = \frac{1}{x^2y^2}$ is an IF of $M\,dx + N\,dy = 0$.
- Thus, $\left(\frac{1}{y} - \frac{2}{x}\right)dx + \left(\frac{3}{y} - \frac{x^2}{y^2}\right)dy = 0$ is an exact DE.
- $\therefore \; \exists\, F(x,y)$ such that

$$\frac{\partial F}{\partial x} = \frac{1}{y} - \frac{2}{x}, \qquad \frac{\partial F}{\partial y} = \frac{3}{y} - \frac{x^2}{y^2} \tag{3}$$

- The GS is $F(x,y) = \dfrac{x}{y} + \ln\left(\dfrac{y^3}{x^2}\right) = C$ (HOMEWORK).



Observations:

Solve $(x^2 + y^2)\,dx + xy\,dy = 0$. (HOMEWORK).

Note: If $Mx + Ny = 0$, then $\dfrac{M}{N} = -\dfrac{y}{x}$.

$$\therefore \quad M\,dx + N\,dy = 0 \;\Rightarrow\; \frac{M}{N} = -\frac{dy}{dx}$$

$$\therefore \quad \frac{dy}{dx} = \frac{y}{x}$$

$$\Rightarrow \quad y\,dx - x\,dy = 0$$

$$\Rightarrow \quad \frac{y\,dx - x\,dy}{y^2} = 0$$

$$\Rightarrow \quad d\left(\frac{x}{y}\right) = 0$$

$$\Rightarrow \quad \frac{x}{y} = k \;\Rightarrow\; x = ky \quad \text{is the GS of the DE.}$$



Observations:

If $Mx - Ny = 0$, then $\dfrac{M}{N} = \dfrac{y}{x}$.

$$\therefore \quad M\,dx + N\,dy = 0 \;\Rightarrow\; \frac{M}{N} = -\frac{dy}{dx}$$

$$\therefore \quad -\frac{dy}{dx} = \frac{y}{x} \;\Rightarrow\; y\,dx + x\,dy = 0$$

$$\Rightarrow \quad xy = C \quad \text{is the GG of the DE.}$$

Result: If $Mx - Ny \neq 0$ and $M\,dx + N\,dy = 0$ can be written as

$$f_1(xy)\,y\,dx + f_2(xy)\,x\,dy = 0, \tag{4}$$

then $\dfrac{1}{Mx - Ny}$ is an IF.



Sketch of proof:

- From (2),

$$\begin{aligned}
\frac{M\,dx + N\,dy}{Mx - Ny} &= \frac{1}{2} \cdot \frac{Mx + Ny}{Mx - Ny}\, d(\ln xy) + \frac{1}{2}\, d\left(\ln \frac{x}{y}\right) \\
&= \frac{1}{2} \cdot \frac{f_1(xy)\, xy + f_2(xy)\, xy}{f_1(xy)\, xy - f_2(xy)\, xy}\, d(\ln xy) + \frac{1}{2}\, d\left(\ln \frac{x}{y}\right) \\
&= G(xy)\, d(\ln xy) + \frac{1}{2}\, d\left(\ln \frac{x}{y}\right) \qquad \text{(say)} \\
&= H(\ln xy)\, d(\ln xy) + \frac{1}{2}\, d\left(\ln \frac{x}{y}\right)
\end{aligned}$$

  which is an exact differential.

- Thus, $\dfrac{1}{Mx - Ny}$ is an IF in this case.

